



HAMDI JARBOUI

Senior Python Engineer · AI Agents · DevOps

jarbouihamdi9@gmail.com · +216 98 135 227 · Sfax, Tunisia
linkedin.com/in/hamdi-jarboui · github.com/hamdiJarboui

PROFILE

Python software engineer with 4+ years delivering production systems end-to-end — architecture through deployment — across full-stack development, DevOps, and embedded automotive systems. Currently leads the design of AI agent-based automation at KPIT, embedding LLM-driven logic into live engineering workflows and owning CI/CD from commit to production. Works directly with customers to gather requirements and present technical solutions, and is recognised internally as a technical mentor: delivered a generative AI workshop to a full 100-engineer development department and has trained developers, testers, and students in 20-person cohorts across 10–30 hour programmes since 2021. Seeking a senior engineering role where deep Python expertise and AI automation experience drive team-wide impact.

TECHNICAL SKILLS

Python Development: Full-stack Python across web, desktop (PyQt6), embedded, data, and AI applications; back-end services, REST APIs, internal tooling, automation

Generative AI: LLM integration and application development, prompt engineering, context design, output evaluation, GenAI-assisted development and testing workflows

AI Agents: Agent architecture and orchestration, tool/function calling, multi-step task automation, agent-driven engineering workflows in production

Machine Learning: Machine learning, deep learning, computer vision, model training and evaluation

DevOps & CI/CD: Jenkins, Zuul, GitLab CI/CD, Nexus, Artifactory, containerization, pipeline design from commit to production

Test Automation: pytest, Robot Framework, TestGuide, ECU validation, CI and ECU test pipeline management

Data & Dashboards: Grafana, Kibana, Prometheus, Plotly Dash, custom-built dashboards, data analysis and visualization

Cloud & Platforms: AWS, Linux, Git / GitHub / GitLab

Databases: PostgreSQL, MySQL, MongoDB, Oracle, SQLite3

PROFESSIONAL EXPERIENCE

Senior Python Developer & DevOps Engineer — AI Agents

07/2024 - Present

KPIT Technologies

Sfax, Tunisia

- Own the design and delivery of AI agent-based automation systems in Python, setting technical direction for how the team automates repetitive engineering work.
- Architect full-stack Python applications end-to-end — back-end services and APIs through to internal tooling interfaces — making key design decisions independently.
- Own CI/CD pipelines (Jenkins, Zuul, GitLab CI/CD) from commit to production, with artifact management via Nexus and Artifactory and test orchestration through TestGuide and Jenkins.
- Partner with cross-functional teams on containerization, cloud deployment (AWS), and monitoring, improving release reliability beyond individual project scope.
- Build Python dashboards and data-visualization tooling adopted by engineering leads for day-to-day decisions.
- Work directly with customers to gather requirements, present technical solutions, and align deliverables — translating between engineering detail and client expectations.
- Delivered a generative AI workshop to the entire development department (~100 engineers), plus recurring hands-on sessions training developers and QA testers in cohorts of ~20 on Python and Python-based test automation.

Primatec Engineering

Sfax, Tunisia

- Became the reference engineer for Python ECU flashing and configuration tooling, enabling faster embedded systems validation.
- Led automation of build, test, and deployment pipelines with Jenkins, removing manual steps from the team's release process.
- Built full-stack Python tools and dashboards adopted across engineering teams for data analysis and reporting.
- Prototyped automation concepts that became the foundation for AI agent-driven workflows now in production at KPIT.
- Worked directly with customers to gather requirements, present technical solutions, and align deliverables with client expectations.
- Ran hands-on workshops training developers and testers in ~20-person cohorts on Python testing (pytest, Robot Framework).

TECHNICAL TRAINING & INSTRUCTION

5 years training engineers and students in Python, generative AI, and agentic AI — across 2 engineering organisations and 4 institutions. 20-person cohorts, 10–30 hour programmes, 100% PCEP certification pass rate, and students placing 3 times at national AI competitions.

Lead Instructor — Python, Generative AI & Computer Vision · *2S Training and Consulting*

2021 – Present

Own curriculum design and delivery end-to-end across Python, generative AI, prompt engineering, agentic AI, data analysis, and computer vision. Run 20-person cohorts in 10–30 hour programmes, translating LLM and agent architecture into hands-on practice. Teaching these topics since 2021, ahead of mainstream adoption.

Python & Applied AI Instructor · *ENETCOM Engineering School, Sfax*

2024 – Present

Selected to teach Python, generative AI, prompt engineering, and agentic AI to engineering students at a national engineering school. Coached student teams to 3 national-level AI competition wins.

Python & Generative AI Instructor · *ISB — International School of Business*

2024 – Present

Deliver Python, generative AI, and prompt engineering to a non-engineering business audience — translating technical material for stakeholders without a coding background.

Python Instructor · *Smart Skills Academy*

2022 – 2023

Built a full PCEP/PCAP-aligned curriculum from scratch and supervised student projects in Python, data analysis, and introductory AI.

EDUCATION**Engineering Degree, Industrial Computer Engineering***National School of Electronics and Telecommunications (ENETCOM), Sfax · 2019 – 2024***Preparatory Cycle, Physics & Chemistry***Faculty of Sciences, Sfax · 2017 – 2019***CERTIFICATIONS**

- Python Institute — PCEP (Certified Entry-Level Python Programmer) · PCAP (Certified Associate in Python Programming)
- IBM — Applied Data Science with Python (Level 2) · IBM — Deep Learning (Level 2): TensorFlow, GPU-accelerated training
- University of Washington (Coursera) — Machine Learning Specialization
- Microsoft MTA — Programming Using Python & Java · Certiprof — Scrum Foundation (SFPC)

ACHIEVEMENTS

- 100% PCEP certification pass rate — every engineer trained through the in-company Python programme achieved Python Institute PCEP certification
- Trained 100+ engineers in a single department-wide generative AI workshop; ongoing 20-person cohorts across 10–30 hour programmes since 2022

LANGUAGES**Arabic:** Native / bilingual proficiency**French:** Full professional proficiency**English:** Professional working proficiency